ASSIGNMENT 3:

Module 4: Functions & Modules in Python

Task 1: Calculate Factorial Using a Function

```python
#Calculate Factorial Using Function

def factorial(num):  2 usages
    if num == 1:
        return 1
    else:
        return num * factorial(num-1)

num = int(input("Enter a number: "))
print(f"Factorial of {num} is: {factorial(num)}")
```

Run 9.16 simple banking application Task 3.1

```
/Users/kirandeepsohi/PyCharmMiscProject/.venv/bin/python /Users/kira
Enter a number: 10
Factorial of 10 is: 3628800

Process finished with exit code 0
```

Task 2: Using the Math Module for Calculations

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```python
#using the math module for calculations
#sin(x, /)- Return the sine of x (measured in radians).
#log(...)- Return the logarithm of x to the given base.
#sqrt(x, /)-Return the square root of x.
#to check these i have used
# help(sin)
# help(log)
# help(sqrt)

from math import sin, log, sqrt
num = int(input("Enter a number: "))
square_root = sqrt(num)
Logarithm = log(num)
Sine = sin(num)
print(f"Square root:{square_root}")
print(f"Logarithm: {Logarithm}")
print(f"Sine: {Sine}")
```

9.16 simple banking application | Task 3.2

```
/Users/kirandeepsohi/PyCharmMiscProject/.venv/bin/python /Users/kirande
Enter a number: 30
Square root:5.477225575051661
Logarithm: 3.4011973816621555
Sine: -0.9880316240928618

Process finished with exit code 0
```

GitHub - kirandeepkaursohi-maker/Python-Basics